

Gas Laws 1

Essential Pre-Uni Physics G2.1

Subject & topics: Physics | Thermal | Gases **Stage & difficulty:** A Level P2

Physical constants which may be necessary to answer the problems on this page can be found within the hint tabs.

Don't forget that one mole of gas contains 6.02×10^{23} molecules, and that the mass of this amount is called the 'molar mass'.

What is the volume of a mole of gas at atmospheric pressure (1.01×10^5 Pa) and at 20°C ? Give your answer to two significant figures.

Question deck:

STEM SMART Physics 28 - Gases & Kinetic Theory

Gas Laws 3

Essential Pre-Uni Physics G2.3

Subject & topics: Physics | Thermal | Gases **Stage & difficulty:** A Level P2

Physical constants which may be necessary to answer the problems on this page can be found within the hint tabs.

Don't forget that one mole of gas contains 6.02×10^{23} molecules, and that the mass of this amount is called the 'molar mass'.

How many molecules of gas do you need in a 100 cm^3 cylinder to exert a pressure of $1.0 \times 10^8 \text{ Pa}$ at a temperature of 800°C ? Give your answer to 2 significant figures.

Question deck:

STEM SMART Physics 28 - Gases & Kinetic Theory

Gas Laws 5

Essential Pre-Uni Physics G2.5

Subject & topics: Physics | Thermal | Gases **Stage & difficulty:** GCSE P3, A Level P2

Physical constants which may be necessary to answer the problems on this page can be found within the hint tabs.

Don't forget that one mole of gas contains 6.02×10^{23} molecules, and that the mass of this amount is called the 'molar mass'.

A tyre contains 800 cm^3 of air at a pressure of about $5.0 \times 10^5 \text{ Pa}$ at 9.0°C . After a cycle ride, the volume is 810 cm^3 and the temperature is now 25°C . Assuming that none of the gas has leaked, what is the new pressure?

Question deck:

STEM SMART Physics 28 - Gases & Kinetic Theory

Gas Laws 6

Essential Pre-Uni Physics G2.6

Subject & topics: Physics | Thermal | Gases **Stage & difficulty:** A Level C2

Physical constants which may be necessary to answer the problems on this page can be found within the hint tabs.

Don't forget that one mole of gas contains 6.02×10^{23} molecules, and that the mass of this amount is called the 'molar mass'.

A tyre contains 800 cm^3 of air at a pressure of about $5.0 \times 10^5 \text{ Pa}$ at 9°C . After a cycle ride, the volume is 760 cm^3 , the temperature is now 25°C , and the pressure is $4.0 \times 10^5 \text{ Pa}$. What percentage of the gas molecules have leaked out?

Question deck:

STEM SMART Physics 28 - Gases & Kinetic Theory

Gas Laws, Density and Kinetic Energy 2

Linking Concepts in Pre-Uni Physics 32.2

Subject & topics: Physics | Thermal | Gases **Stage & difficulty:** A Level P3

Element	Molar mass / g mol^{-1}
H	1
S	32
O	16

What is the density of a sulfuric acid gas cloud on Venus if the temperature is 467°C and the pressure is 9308 kPa ? The chemical formula for sulfuric acid is H_2SO_4 .

Gas Laws, Density and Kinetic Energy 6

Linking Concepts in Pre-Uni Physics 32.6

Subject & topics: Physics | Thermal | Gases

Stage & difficulty: A Level P3

Calculate the mean kinetic energy of molecules in a gas at 15°C .

Question deck:

STEM SMART Physics 28 - Gases & Kinetic Theory

Gas Laws, Density and Kinetic Energy 8

Linking Concepts in Pre-Uni Physics 32.8

Subject & topics: Physics | Thermal | Gases Stage & difficulty: A Level P3

Within a gas mixture at equilibrium, the mean kinetic energy of each type of molecule is the same. This is because the temperature is uniform. In a mixture of helium (molecular mass $m = 4.00 \text{ u}$) and nitrogen ($m = 28.0 \text{ u}$),

Part A

Which molecules move faster

state which molecules move faster.

- ☐ Helium
- ☐ They both move at the same speed.
- ☐ Nitrogen

Part B

Ratio

calculate the ratio of their mean square speeds $\overline{c_{\text{helium}}^2} / \overline{c_{\text{nitrogen}}^2}$.

Question deck:

[STEM SMART Physics 28 - Gases & Kinetic Theory](#)

Photon Gas

100 Challenge Problems in Pre-Uni Physics Q88

Subject & topics: Physics | Thermal | Gases **Stage & difficulty:** A Level C3



Make sure you are familiar with the kinetic theory derivation for an ideal gas before attempting this question.

The concept of a **photon gas** is useful in the field of thermodynamics.

Instead of a box containing an ideal gas, consider a box of volume V containing a large constant number of photons N . Assume that all photons have the same energy E . (This assumption simplifies the algebra, but in reality the photon energies will follow a certain distribution.)

In this question, we will be adapting the kinetic theory derivation for an ideal gas to derive the pressure of a photon gas.

Part A

Force

A photon has momentum p which is related to its energy through $p = E/c$.

First consider collisions with a wall perpendicular to the x -axis. The components of the photon's momentum and speed in this direction are related via $\frac{p_x}{p} = \frac{c_x}{c}$.

The side length of the box parallel to the x -direction is L_x . The walls of the box are perfectly reflective.

Find an expression for the force due to one photon repeatedly colliding with the wall in terms of p , c_x , c and L_x .

The following symbols may be useful: L_x , c , p , p_x

Part B

Mean square speed

In a similar way to the derivation for an ideal gas, we define a mean square velocity component

$$\overline{c_x^2} = \frac{c_{x,1}^2 + c_{x,2}^2 + c_{x,3}^2 + \dots + c_{x,N}^2}{N}$$

where $c_{x,i}$ is the x -component of the velocity of the i^{th} photon.

Hence show that the pressure due to all N photons is given by $P = k\overline{c_x^2}$ and give the expression for k in terms of p , c , N and V .

The following symbols may be useful: N , V , c , k , p

Part C

Pressure and internal energy

The total internal energy U is the sum of all photon energies in the box.

Show that the pressure in the box is $P = \alpha U/V$ and give the value of α .

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Question deck:

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